

NATIONAL BUREAU OF STATISTICS

References and Proof

Sources and reproducibility

Nigeria Music Analytics System

Delivery 130 · 129 artists · Q1 2019 – Q3 2026

1 September 2026

References and Proof — DELIVERY130

Data providers

Provider	Role	What it supplies
Chartmetric	Primary	Artist identity, Spotify/YouTube/Deezer audience levels
Soundcharts	Primary	Audience levels, listener geography by country and city

Both are commercial music-data providers. Every observed figure in this package traces to one of them through `04_Datasets/Daily_Metric_Observations.csv`, which carries `source_endpoint` and `source_field` on every row.

What is NOT sourced from a provider

The constants below are assumptions or estimates taken from outside the providers. They are listed in full so a reviewer can challenge any of them:

Constant	Value	Class	Source
<code>STREAMS_PER_LISTENER_MONTH</code>	3.5	EST	Industry proxy. Not derived from Nigerian data.
<code>SPOTIFY_PER_STREAM</code>	0.004	EST	Industry average (Ditto Music 2026, Chartlex 2026).
<code>YOUTUBE_PER_VIEW</code>	0.004	EST	Industry average (Hootsuite 2025).
<code>DEEZER_PER_STREAM</code>	0.004	EST	Industry average.
<code>DEEZER_STREAMS_PER_FAN_MONTH</code>	2.0	EST	Industry proxy.
<code>VIEWS_PER_SUBSCRIBER_MONTH</code>	13.527	EST	Calibrated: OLS on the AGGREGATE observed ratio across the 19 fully-observed quarters Q4 2021 - Q2 2026, $R^2 = 0.782$. Replaces the first submission's unsourced flat 15.0.
<code>COUNTER_RESTATEMENT_FACTOR</code>	50.0	ASM	Set from the observed distribution: 44 of 2,168 artist-quarters exceed 50x, headed by a single day of 826,001,508 views against a 351,682 median day. No genuine quarter approaches it.
<code>VIEWS_PER_LISTENER_MONTH</code>	12.48	EST	Calibrated: aggregate $\text{sum}(\text{views})/\text{sum}(\text{listeners})/3$ over the 1,703 artist-quarters carrying both series.
<code>MIN_OBSERVED_SPAN_COVERAGE</code>	0.9	ASM	Set from observed coverage: 19 quarters span 97.8-100%, Q3 2021 spans 8.7% and Q3 2026 (unfinished) 42.4%. The threshold separates those two from every adequately observed quarter.
<code>UNMEASURED_UPLIFT_RATE</code>	0.3	ASM	Assumed from Spotify's approximate market share. Verified against the delivered file, which reproduces at exactly this value.
<code>NAIRA_PER_USD</code>	1500	ASM	Single assumed rate.
<code>TRACKS_PER_QUARTER</code>	2	ASM	Assumption.
<code>AVG_PRODUCTION_COST_NGN</code>	750000	ASM	NigerianInformer 2025 (secondary).

Constant	Value	Class	Source
AVG_DISTRIBUTION_COST_NGN	15000	ASM	Bliss hype 2026 (secondary).
AVG_PROMOTION_COST_NGN	250000	ASM	TaGetMedia 2025 (secondary).
AVG_HOSTING_COST_QUARTERLY_NGN	50000	ASM	Industry estimate (secondary).

Proof of reproducibility

Every published figure can be recomputed from the published columns:

- $\text{spotify_revenue_usd} = \text{spotify_monthly_listeners} \times 3.5 \times 3 \times 0.004$
- $\text{youtube_revenue_usd} = \text{youtube_actual_views} \times 0.004$
- $\text{deezer_revenue_usd} = \text{deezer_fans} \times 2.0 \times 3 \times 0.004$
- $\text{other_platforms_revenue_usd} = \text{spotify_revenue_usd} \times 0.30$
- $\text{gross_streaming_revenue_usd}$ = the four above, summed
- $\text{gross_streaming_revenue_ngn} = \text{gross_streaming_revenue_usd} \times 1500$

These identities hold on **all 3,798 rows** of `Gross_Streaming_Revenue.csv`. A reviewer needs nothing from this system to check them.